claude-windows / 144a31b7-7962-4fee-9d6f-8a7854daa63f

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\144a31b7-7962-4fee-9d6f-8a7854daa63f\subagents\workflows\wf_1c5f1f5c-9ba\agent-adf6631ff8e5a5a2e.jsonl`
- SHA-256: `f0365eeca707d2f5e5f6e7b1688a820642ef18a3bf969f3111aa08deed2c1279`
- Source modified: `2026-06-20T08:01:32+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_1c5f1f5c-9ba`
- session_id: `144a31b7-7962-4fee-9d6f-8a7854daa63f`

Transcript

1. user (2026-06-20T08:00:25.441Z)

Extract the EXACT structure a tracked design/project doc must follow in this project.
Read: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PROJECT_DOC_TEMPLATE.md (the canonical template), and skim docs/RALLY_QUALITY_RESEARCH.md and docs/PROMINENT_COURT_DETECTION.md as real examples of the template in use.
Also read C:\Users\avidu\Projects\khelsutra\docs\REQUEST_ACCESS_FORM.md and C:\Users\avidu\Projects\khelsutra\docs\PORTABILITY.md to match the khelsutra repo's doc voice.
Report: required_sections (in order), header_fields (the status-header fields like Status/Owner/Decisions-locked/etc.), the progress_tracker_format, the definition-of-done format, and style_notes (numbering convention, tone). Quote path:line. Return the schema.

2. assistant (2026-06-20T08:00:33.088Z)

I'll read the canonical template and the example docs, plus the khelsutra docs for voice matching.

3. user (2026-06-20T08:00:34.089Z)

```
<!--
PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.

A "tracked project doc" is how this repo carries substantial design/project work: it can be
ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables, then
archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
docs/GOLDEN_REGRESSION_FIXTURES.md.

Rules of thumb:
- §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable PR.
- Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs sign-off),
  or [design] (proposed). The doc reflects real shipped state, never aspirational.
- It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
- Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model for
  research- or security/privacy-heavy work). Delete sections that do not apply.
- Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state work).
-->

# <Project / Plan Title>

> **Status:** `DRAFT` ? `<owner-gated? | in progress>` · **Owner:** `<name>` ·
```

```
**Created:** `<YYYY-MM-DD>` · **Last updated:** `<YYYY-MM-DD>`
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
> **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
> **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not legal/financial advice where applicable.

---

## 0. TL;DR
<2?5 sentences: the goal, the approach, and the single most important caveat. Call out any finding that is already actionable on its own.>

## 1. Problem & goal
<Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a reader must read to get current.>

## 2. Decisions locked
<Table ? capture answered questions so they are not relitigated.>

| # | Decision | Source / date | Implication |
|---|----------|---------------|-------------|
| D1 | ? | ? | ? |

## 3. Foundation ? research / prior art  *(OPTIONAL)*
<Include when the plan rests on external facts: cited research, legal/compliance, benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this creates.>

## 4. Design / architecture
<The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing modules/code this builds on so it is clear what is new vs reused.>

## 5. Threat model / risk table  *(OPTIONAL ? security / privacy / compliance work)*
<Who/what is protected vs NOT; residual risks stated plainly.>

## 6. Honest limits ? what this does NOT do
<The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>

## 7. Deliverables & progress tracker   ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

| ID | Deliverable | Depends on | Gated? | Status | PR |
|----|-------------|-----------|--------|--------|----|
| P0 | ? | ? | No | ? | ? |

## 8. Open questions ? owner / external  *(split blocking-gates from nice-to-knows; name who decides)*
1. ?

## 9. Definition of done
<Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
- [ ] ?

## 10. References
**Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`. **External:** `<cited sources>`.

---

### Changelog
- `<YYYY-MM-DD>` ? `<what changed>`.
```

75

4. user (2026-06-20T08:00:35.502Z)

1 # Prominent-Court Detection ? kill background-court false positives (multicourt)
2
3 > **Status:** `DRAFT` ? design + experiment **owner-gated for any default flip** · **Owner:** Avi · **Created:** 2026-06-19 · **Last updated:** 2026-06-19
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5 > **Tracking anchors:** §8 progress tracker is the source of truth; pointer in memory `current-state`.
6 > **Relation to existing docs:** attacks gap **G2** ("precision ? local over-fires, worst on multicourt") from [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md); it is the **shuttle-side** counterpart of the person-side **Gate B / court mask** in [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md). Complements the held-shuttle **`shuttle_player_colocation`** (G3) ? court-localization (G2) and held-shuttle (G3) are independent precision levers.
7 > **Honesty note:** claims are tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). Every signal lands **flag-gated, default-OFF**, promoted only on measured lift per [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) + [[decision-rigor-norm]].
8
9 ---
10
11 ## 0. TL;DR
12 In **multicourt** footage the camera sees several courts. The shuttle detector (WASB) tracks *any* shuttle in frame, so a rally on a **background / adjacent court** becomes a predicted rally on the **main match** ? a false positive. The fix: identify the **prominent court** (the foreground court that occupies the majority of the frame, where the match being filmed is played) and **keep only rallies whose shuttle lives inside it**. Mechanically this re-defines the shuttle signal from `(x, y, visible)` to `(x, y, visible-AND-inside-the-prominent-court)` *before* windowing ? so a background-court shuttle reads as "not there" and never forms a rally. A flat 2D floor polygon is the MVP; a **pseudo-3D play-volume** (court floor + the air column above it) is the refinement so a high clear over the prominent court isn't wrongly rejected. Lands as a **default-OFF experiment**; the flip is gated on a Launch Report ([[launch-report-convention]]).
13
14 ## 1. Problem & motivating example `[verified]`
15 - **Measured gap (G2):** local over-fires ~2× vs Gemini, and the over-fire is **worst on multicourt** ? even among long (>7s) rallies, multicourt over-fires **1.58×** (background-court games are *real* long rallies, just on the wrong court). See `RALLY_DETECTION_GAP_CLOSURE.md` §3 + the P7 long-rally lens.
16 - **Concrete anchor ? GX010142 rally #1 (2026-06-19).** Owner observation: *"the first rally is just a shuttle flying in the non-prominent court? while the prominent-court players have not started."* Confirmed from the cached WASB trajectory (`output/wasb/GX010142_1080p__*_wasb.csv`): rally #1 (`0.72?3.65s`, 2.94s) has a shuttle centroid at **meanX ? 1616** on a 1920-wide frame (?84% to the right) ? spatially apart from the central cluster where most rallies live (meanX ? 1000?1300). Several other suspicious rallies (#3, #9, #12, #16) cluster in the same far-right band ? consistent with a **right-side adjacent court**.
17 - **Why a 1D threshold is not enough `[verified]`:** the courts do not separate cleanly on X or Y alone (rally Y-centroids overlap 250?620 px across courts; the far-right band mixes with foreground play). The courts are **2D regions** under perspective ? so we need the court *geometry*, not a coordinate cutoff. This is the owner's "pseudo-3D / court model" intuition, validated by the data.
18
19 ## 2. Goal & non-goals
20 - **Goal:** on multicourt clips, drop rallies whose shuttle is outside the prominent court, **without losing real prominent-court rallies** ? measured per-video, no regression on single-court clips (where it must be a no-op or a do-no-harm).
21 - **Non-goals (v1):** multi-camera; tracking players across courts; full court-line homography for scoring; anything biometric. Single-court footage is explicitly a **no-op** (one court ? everything is "prominent").
22
23 ## 3. Decisions locked

#	Decision	Rationale
D1	The prominent court is a **2D region (polygon) in image space**, refined to a **pseudo-3D play volume**	1D thresholds don't separate courts under perspective (§1)
D2	Gate the **shuttle trajectory** by the region **before windowing** (re-define visibility)	Stops a background rally at the source ? no window ever forms there. Reuses `point_in_polygon`
D3	**Default-OFF, flag-gated; single-court = no-op**	Zero-regression rollout ([[weak-signals-retained]]); no behaviour change unless a polygon is supplied + the flag set
D4	Reuse the existing **`court_polygon`** config + `point_in_polygon`; do NOT invent a new geometry stack	`FusionConfig.court_polygon` + `fusion_features.point_in_polygon` already exist `[verified]`
D5	The prominent court can be **supplied (config) OR auto-derived**; auto-derivation is its own measured step	Config polygon is the cheapest first experiment; auto is the product path

4. Design

4.1 The signal change (the core experiment)
Today the windower consumes the raw WASB trajectory: a per-frame `(Frame, Visibility, X, Y)`. The change is a **pre-windowing filter**:

```
for each frame f:  if shuttle_visible(f) and NOT inside_prominent_court(X_f, Y_f):  mark f as NOT visible
```

A background-court shuttle is then indistinguishable from "no shuttle" ? the velocity windower never opens a rally there. This is purely **subtractive** on the shuttle stream (it can only remove visibility, never add it), so on a correctly-drawn prominent court it cannot manufacture rallies ? only suppress out-of-court ones. (Mirror of `rally_gate`'s "do-no-harm" discipline.)

4.2 Identifying the prominent court (four routes, cheapest first)
1. **Configured polygon** (MVP, `[design]?buildable now`): a normalized `[[x,y],?]` court polygon per camera/clip, exactly the existing `FusionConfig.court_polygon`. Draw it once from the dense shuttle/player cluster. Good enough to validate the whole concept on GX010142 + the multicourt golden clips.
2. **Player-location auto-derivation** (`[design]`): run `PersonDetector.detections_per_frame` `[verified] exists`, take the **largest, most-central, most-persistent** player cluster (the foreground players are bigger boxes lower in frame), and fit the prominent-court polygon to where *those* players' feet live. "Prominent = where the match's players are." This is why the owner asked to **extract person features** ? players anchor the court.
3. **Shuttle-density auto-derivation** (`[design]`): the prominent court is the densest sustained shuttle-activity region; cluster the visible trajectory and take the dominant mode. Person-free fallback.
4. **Court-line detection** (`[design]`, heaviest): Hough/segmentation court lines ? the largest court quad. Most accurate, most work; deferred.

v1 ships route 1 (config) + measures; route 2 (player-anchored) is the product upgrade.

4.3 Pseudo-3D play volume (the refinement)
A flat floor polygon rejects a shuttle at the top of a high clear *over the prominent court* (it projects above the floor quad). The pseudo-3D model treats the prominent court as a **floor quad + a vertical air column**: accept a shuttle if its image point lies within the floor polygon **expanded upward** by a height envelope (the max plausible shuttle height projected through the camera). Practically: the floor quad plus an upward `pad` (perspective-aware ? larger near the net/far baseline). Start with a generous fixed upward pad (cheap, MVP); upgrade to a per-camera height envelope only if measurement shows real rallies being clipped. This keeps the floor tight (rejects adjacent-court floor play) while not clipping legitimate high shuttles.

4.4 Reuse map `[verified]`
| Need | Existing code |

point-in-region test	`backend/pipeline/detectors/fusion_features.py::point_in_polygon`
normalized court polygon config	`backend/config/models.py::FusionConfig.court_polygon` (today: person/Gate-B only)
player boxes per frame	`backend/pipeline/detectors/person_detector.py::PersonDetector.detections_per_frame`
trajectory parse + windowing	`backend/pipeline/detectors/tracknet_runner.parse_trajectory_csv`, `backend/eval/windowing.py`
per-video scoring	`backend/eval/rally_seg_eval.py` (+ the `--stratify` multicourt split, #220)

New code is small: a `prominent_court_gate(points, polygon, height_pad)` that filters trajectory points, threaded into the windowing input behind a default-OFF flag.

4.5 Extension ? deterministic rally-END via shuttle-floor contact (owner note, 2026-06-19) `[design]`
The same court geometry that localizes the prominent court also unlocks a **deterministic rally-END signal** ? the cheapest, most certain way to know a point is over: **the shuttle hit the floor.** This attacks **G3** (rally-END over-extension), where today's shuttle-velocity windowing runs the window long because "shuttle still moving ? rally over."

- **Introduce two geometric primitives:**
  - **The white court lines.** A badminton court is bounded by **white, straight lines forming a rectangle** (with the perspective trapezoid in image space). Detect them ? Hough lines / line-segment detection filtered to white, long, court-consistent segments ? to recover the **court quad** (which also *is* the prominent-court floor polygon of §4.2 route 4; one detector serves both).
  - **The floor plane.** The court lines define the **floor** (the plane the players stand on). The shuttle in flight lives *above* this plane; a shuttle *at* the plane (within the trapezoid, or just outside it) is **on the floor**.
- **The rally-END rule (deterministic):** a rally ends when the shuttle makes **floor contact** ? its trajectory descends to and **stops/rests at the floor level** (a near-zero-velocity terminus at the court-line plane), OR a shuttle that goes to the floor **outside the prominent court** = an out-of-bounds landing (also point-over). Unlike velocity heuristics, a shuttle resting on the floor is **unambiguous** ? the point is over.
- **Why it's strong + cheap:** no new model, no person stack ? it reuses the WASB trajectory (which already gives the shuttle's descending `(x, y)` + the `Visibility` drop when it settles) plus the court-line geometry. It is the **shuttle-side** rally-END signal, complementary to the **player-state** rally-END (player kinematics / held-shuttle) in `RALLY_DETECTION_GAP_CLOSURE.md` §4 Lever A ? fuse both for robustness (a shuttle can settle off-camera; a player can pick up early).
- **Subtlety to measure:** distinguish a true floor landing (point over) from a shuttle that **dips low but is played back up** (a tight net shot / low retrieve). The court-line plane + a brief **dwell** at floor level (the shuttle rests, doesn't re-accelerate up) is the discriminator; tune the dwell window on golden clips.

This is tracked as **C8** below and cross-links gap-closure **G3**. It shares the court-line detector with the prominent-court polygon (§4.2 route 4), so the two land together once court-line detection exists.

5. Experiment plan (measurement)
Land the gate flag-gated, then measure on the **cached trajectories** (CPU, no GPU/re-detect):
1. **GX010142 sanity:** draw a prominent-court polygon; confirm **rally #1 disappears** and the central rallies survive. The litmus the whole idea is born from.
2. **Multicourt golden** (`mahadevpura_2, GX010128, Badminton_BXH_2, Boxhill_Doubles`): ? rally-count, and once those clips are golden-labelled, ? precision / ? recall vs golden via `rally_seg_eval --stratify` (single vs multicourt). The win condition: multicourt precision up, **recall flat** (we drop FPs, not real rallies).
3. **Single-court no-op:** on single-court clips with no polygon (or a full-frame polygon) the gate is a verified no-op ? assert zero change.
4. **Dual-eval discipline** ([[eval-dual-set-regression]]): no per-video regression on either eval set.

Promotion to default-ON is a **separate, owner-gated** decision with a Launch Report (both rulers + the over-seg guard), never bundled with the experiment PR.

6. Risks & limits
- **Polygon is camera-dependent.** A hand-drawn polygon doesn't generalize across setups ? route 2/3 (auto-derivation) is the real product path; v1 config-polygon is for *validation*.
- **Shuttle height** (§4.3): too-tight a floor clips high clears; the pseudo-3D pad mitigates, measurement tunes it.
- **The prominent court can change** (camera re-aim mid-clip): out of scope v1 (assume static per clip).
- **Not a recall fix.** This only removes out-of-court FPs; it does nothing for missed in-court rallies (G1) ? and must be proven not to *cost* recall.

7. Open questions (owner / data)
1. Auto-derivation anchor: **player-location** (route 2) vs **shuttle-density** (route 3) as the default once validated? *(decides: data, post-experiment)*
2. Height-envelope: fixed upward pad vs per-camera perspective model ? is the fixed pad enough? *(decides: measurement)*
3. Do we expose the prominent court as a **one-time per-camera setup** (draw-once UI) or always auto? *(decides: owner/product)*

8. Deliverables & progress tracker ? source of truth
Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.

ID	Deliverable	Depends on	Gated?	Status	PR
C0	This design doc	?	No	?	this PR
C1	`prominent_court_gate(points, polygon, height_pad)` + unit tests (pure, point-in-polygon)	?	No (default-OFF)	?	?
C2	Thread the gate into the windowing input behind a default-OFF flag (`court_polygon` + `gate_shuttle_to_court`)	C1	No (default-OFF)	?	?
C3	GX010142 litmus: rally #1 dropped, central rallies kept (measured)	C2	No	?	?
C4	Extract person features (`detections_per_frame`) on GX010142 ? player-cluster = prominent court; measure `shuttle_player_colocation` too	?	No	?	?
C5	Multicourt-golden measurement (? count now; ? precision/recall vs golden once labelled)	C2, labels	? owner labels	?	?
C6	Auto-derive the polygon (route 2 player-anchored)	C4	? owner	?	?
C7	Launch decision (default-ON) with a Launch Report	C3,C5	? owner	?	?
C8	**Court-line / floor detection** ? (a) auto court polygon, (b) **deterministic shuttle-floor-contact rally-END** (§4.5, targets G3)	?	? owner	?	?

9. Definition of done
- [ ] The gate exists, is **default-OFF**, unit-tested, and a verified no-op on single-court / no-polygon.
- [ ] GX010142 rally #1 (and the far-right adjacent-court band) is dropped while the central rallies survive `[measured]`.
- [ ] On the multicourt golden clips, multicourt **precision improves with recall flat** vs golden labels (no per-video regression).
- [ ] A winning configuration is flipped default-ON only via a Launch Report ? flip `Status ? DONE`, archive.

10. References
Code `[verified]`: `fusion_features.py` (`point_in_polygon`, `person_in_court`), `person_detector.py` (`detections_per_frame`), `config/models.py` (`FusionConfig.court_polygon`), `eval/windowing.py`, `eval/rally_seg_eval.py` (`--stratify`).
Docs: [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md) (G2), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (Gate B / court mask), [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md), [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md).

Changelog

- 2026-06-19 ? Created from the owner's GX010142 rally-#1 (background-court FP) observation + the measured multicourt over-fire (G2). Validated the 2D/pseudo-3D need from the cached trajectory; mapped the reuse onto the existing `court_polygon` + `point_in_polygon` machinery.
- 2026-06-19 ? Added §4.5 (owner note): a **deterministic rally-END signal via shuttle-floor contact** ? detect the white court lines ? the floor plane ? a shuttle resting on the floor (or landing out-of-court) ends the rally (targets G3). Shares the court-line detector with the prominent-court polygon; tracked as C8.

5. user (2026-06-20T08:00:35.523Z)

Rally-Detection Quality: Research, Eval Framework & Project Plan

Status: Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the
local?Gemini rally-detection gap,"* **completed when all committed tiers land and a single quality
number is attributed (§7.1).** The empirical + research backbone for closing the gap between the
local, no-cloud rally segmenter and the **Gemini** path. Iterations land as their own measured
PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan they execute against.

How to read this: §1 frames the gap with measured numbers. §2 is the precise, code-grounded
diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we already
have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").
§5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with
falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single
number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable
constraints + cross-links.

> **This doc does NOT duplicate prior design.** It builds on, and cross-links:
> [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),
> [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
> [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),
> [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion framework +
> §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS +
> late-fusion; we *cite, don't re-derive*), [`ABLATION_LAYER.md`](ABLATION_LAYER.md),
> [`EVALUATION.md`](EVALUATION.md), [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
> (TAL-head-first ? VLM), and [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) (corpus growth).

---

1. The gap (measured)

Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video semantically)
is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud) lags
badly. Measured F1 vs golden labels ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md)):

Path	Mahadevpura (clean, single court)	Kushagra (multi-court)
Heuristic velocity windowing (local today)	0.657	**0.157**
Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6)	0.744	0.561
Two-stage WASB + Gemini refine	0.83	?
Gemini wrapper (gemini-flash)	**0.93**	**0.66**

The local heuristic is **footage-fragile**: fine on a clean single court, collapses on multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.

1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)

A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches (Kushagra/Yash June-12; report: `?/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`, generator `scratch/compare_local_vs_gemini.py`). **Gemini is a strong reference here, NOT ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:

	Gemini	Local (WASB no-AI)
rallies / windows (3 videos)	59	12
total "play"	6:28	**17:41**
mean window duration	**6.1 s**	**65 s** (10.6×)
true misses (gemini-only)	?	**0**
merge groups (one local window ? ?2 Gemini rallies)	?	7

Headline: **the local path does not *miss* rallies ? it fails to *segment* them.** One mahadevpura-1 window spanned the entire 174 s clip. Local "play" is 2.7× Gemini's because its mega-windows swallow the dead time between rallies.

---

2. Diagnosis (code-grounded ? read before proposing fixes)

The local pipeline is **two layers** (see [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md)): **(L1) detection** ? per-frame shuttle position (WASB/TrackNet); and **(L2) segmentation** ? turn the trajectory into rally intervals. The gap is almost entirely **L2**.

2.1 L1 (shuttle tracking) is NOT the bottleneck
On the divergence clips WASB reported the shuttle visible in **96?99 % of frames** (mahadevpura-1 98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and dense.

2.2 L2 (segmentation) is the bottleneck ? and the flaw is conceptual
The windowing is `trajectory_to_action_windows` ([`backend/pipeline/detectors/tracknet_runner.py:256`](../backend/pipeline/detectors/tracknet_runner.py)):

- A frame is **"active"** iff the shuttle is **visible** AND its **per-second-normalized speed** exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps / dframes)` ? **already fps-normalized** (? fraction of frame-width per second).
- Active frames within `dead_time_merge_gap` (2.0 s) are **merged** into one window.
- Windows shorter than `min_action_window_duration` (2.0 s) are dropped.

The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress." During dead time the shuttle is still visible and gently moving (being picked up, walked back, knock-ups, slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and because (at the time this was diagnosed) there was **no `max_window_duration` cap and no inactivity/boundary split**, a single window grew until the shuttle finally went invisible ? which, at 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few mega-windows. *(This is the failure W1/R4 fix; both are now default-ON since the R1 milestone, PR #204 ? `max_window_duration=6`, `window_inactivity_end=1.5` in `IndexingConfig`.)*

> **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause was "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes` factor). The real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound. fps matters only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the speed threshold. The `_LOCAL_vs_GEMINI_REPORT.md` has been corrected to match.

2.3 What this implies
The local path **has no rally-STATE signal** ? only "is the shuttle moving." Bridging the gap is fundamentally about giving L2 signals that separate *rally* from *shuttle-merely-moving* (net-crossings, two-sided play, sustained fast exchange, hit sounds) and a segmenter that produces *bounded, well-cut* intervals ? exactly the multi-signal-fusion + learned-segmenter direction the existing docs lay out. This doc makes that concrete and measurable.

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3. What we already have (eval/experiment harness inventory)

The harness is rich and composable ? we **extend** it, not reinvent it. (Modules under `backend/eval/`.)

Scoring & matching
- `metrics.py` ? temporal-IoU interval matching; precision/recall/F1, mean_iou, boundary errors.
- `segmentation_metrics.py` ? **F1@{0.1,0.25,0.5}**, **mAP@tIoU** (0.3?0.7), **segment_count_ratio** (the over/under-segmentation diagnostic). (Edit/segmental score deliberately omitted to date.)

Statistical rigor (small-n)
- `eval_stats.py` ? bootstrap 95 % CIs, paired exact **sign-test**, `paired_delta_ci`, `grouped_folds` (LOGO), `out_of_fold_predictions` (anti-stacking-leakage). The promotion bar = "?F1 CI excludes 0 **AND** sign-test agrees."
- `score_calibration.py` ? Platt + isotonic calibrators, **Brier + ECE**.

Variants, A/B, leaderboard
- `experiment.py` ? `Variant` (declarative recipe: segmenter + config + cue_flags + gates + learned-classifier + threshold/merge_gap/calibrate), `compare()` (LOVO-honest A/B with an "honest" stats block), `compare_unlabeled()` (agreement on unlabeled footage), `CONTROL` (pinned baseline), `record_experiment()` ? `eval_baselines/experiment_leaderboard.json`.
- `ablation.py` ? leave-one-signal-out marginal ?F1 with CI + sign-test + a no-over-claim verdict

vocabulary (`earns_keep` / `suggestive` / `inert` / `structural_protected`).
- `classifier.py` ? dependency-free numpy `LogisticRegression` (the learned scorer; JSON-serializable).

Eval?serve discipline + gates
- `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults so eval and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the golden feature substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but ?0.008 on SERVED** and was reverted ? the canonical eval?serve lesson.)
- `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the served windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues; `regression.py` ? golden-set no-regression baseline with a GT-hash to detect label changes.

Golden substrate
- `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` / `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
- **Corpus today: 15 golden videos** (started at the 6-video label-set hash `50d98ffbc370`, since grown with the GX/Boxhill multicourt clips); more expected via the Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).

Signals available to L2 (built, default-OFF): net-crossing count (`rally_gate.py`, Gate-A), 5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit features (`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` / `_select_for_handoff`. Persisted in `candidate_segments.stats`.

3.1 Known gaps in the harness (what §4 fills)
1. **Over-segmentation is under-surfaced in the default report.** `segment_count_ratio` exists but the merge/split *breakdown* (the very thing that exposed the over-merge) lives only in a scratch script. tIoU-F1 alone is **blind to over-merge** ? a giant window can still "match" via overlap.
2. **No segmental edit-score** (Levenshtein over the predicted/gt label sequence) ? the standard TAS over-segmentation metric.
3. **Per-cue calibration is fitted but never *measured* post-hoc** (no ECE/Brier delta reported).
4. **`out_of_fold_predictions` (anti-leakage stacking) is built but not wired to an arbiter.**
5. **No active-learning loop** ? nothing tells us *which* unlabeled video to label next.
6. **No single "rally-segmentation eval" entrypoint** ? the pieces exist but a one-command `score this segmenter vs golden, honestly` runner would make every iteration cheap.
7. **The local-vs-Gemini agreement track is ad-hoc** (a scratch script), not a first-class *shadow* signal alongside the golden gate.

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4. The strengthened evaluation framework (the "really strong harness")

Design principles (inherited, non-negotiable):
- **Signals ? learned scorer, NO special-casing** ([`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) §1): every new cue is a feature column the scorer learns to weight; never an `if/else` in

the decision path.
- **Promote only on measured lift; default OFF; revert is one flag** ([`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) §6).
- **eval ? serve:** a harness win must be re-confirmed on the *served* windowing before any default flip.
- **Honest small-n stats:** never a bare mean at n?6?16; always CI + paired sign-test (C1).

4.1 Metric set the leaderboard should standardize on
Surface ALL of these in the default `compare()` report (most already computed; the task is to
report them together so over-merge can never hide again):

- **tIoU-F1 @0.5** (primary) **and F1@{0.1,0.25,0.5}** (segmental, TAS-standard).
- **mAP@tIoU** (0.3?0.7) when predictions carry confidences.
- **`segment_count_ratio`** + a **merge-rate / split-rate** breakdown (fraction of GT rallies that
  share a predicted window = under-segmentation; fraction of predicted windows that span ?2 GT =
  the over-merge signal). Promote the scratch `compare_local_vs_gemini.py` matching into a reusable
  `backend/eval/` helper.
- **Segmental edit (Levenshtein) score** ? the canonical over-segmentation metric we currently lack.
- **Boundary timing** (mean start/end error on matched pairs) ? already in `metrics.py`, surface it.
- **Calibration (ECE/Brier)** reported pre/post per-cue calibration (close gap #3).

> **Why:** plain tIoU-F1 rewarded our mega-windows (they "overlap" real rallies). The metric set
> above makes over/under-segmentation *first-class*, so a fix that improves segmentation is
> visible even when tIoU-F1 barely moves ? and vice-versa.

4.2 Splits & significance
- **Leave-one-video-out, grouped by match** (never leave-one-*window*-out ? windows within a video
  are correlated). Keep `grouped_folds`. Note the **RLOOCV bias** caveat for tiny n (see §5.4).
- **Bootstrap CI + paired exact sign-test** on per-video held-out F1 (C1). A result is *promotable*
  only if CI excludes 0 **and** the sign-test agrees, **and** it survives the served-windowing check.
- **Anti-leakage:** per-LOVO-fold threshold tuning + calibration on the TRAIN fold only (already
  enforced in `experiment._calibrate_and_tune`); wire `out_of_fold_predictions` once an arbiter exists.

4.3 The golden corpus + active learning (the highest-leverage lever)
The single biggest quality lever is **more labeled matches** ([`NEXT_STEPS.md`](NEXT_STEPS.md):
"corpus growth = highest leverage"). With ~6 ? ~16 videos:
- Grow via the Roora vendoring flow ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md),
  [`ROORA_LABELING_GUIDELINES.md`](ROORA_LABELING_GUIDELINES.md)) ? court calibration, IAA, the
  3 non-negotiables (ignore dead time, every rally, one-contact = one shot).
- **Active learning:** stop labeling videos at random. Score the *unlabeled* corpus and label the
  videos where the local segmenter is most *uncertain* / most *divergent from Gemini* / most
  diverse (new venue, fps, lighting). §5.3 surveys the methods; the local-vs-Gemini agreement
  signal (§4.4) is a ready proxy for "where do we disagree most ? label that."

4.4 Shadow evaluation: Gemini-agreement as a free, unlimited signal (NOT a gate, NOT training data)
We have exactly one strong reference available on *unlabeled* footage at inference time: **Gemini**.
- `experiment.compare_unlabeled()` already scores agreement (agreed / a-only / b-only) without labels.
- Use it as a **shadow metric**: run any candidate segmenter on a large unlabeled pool, measure agreement-with-Gemini, and watch it as a *trend* ? cheap, unlimited, catches regressions the 6 golden videos can't. **It is a shadow signal only.** Golden labels remain the gate; Gemini is not ground truth (it has its own errors) and ? per the licensing guardrail (§8) ? **its outputs may never become training data for owned weights.**

4.5 One-command rally-segmentation eval (close gap #6)
Wrap the above into a single reusable entrypoint (a future PR ? not built here): given a segmenter variant + the golden manifest, emit the full §4.1 metric set + the honest stats block + a leaderboard row, and (optionally) the §4.4 shadow agreement on an unlabeled pool. Every iteration in §6 is then "run the entrypoint, read the honest delta, promote-or-not."

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5. External research synthesis (cited, adversarially verified)

> Produced by a deep-research run (6 search angles ? 27 sources ? 126 candidate claims ? 25
> 3-vote-adversarially-verified, 23 confirmed / 2 refuted). **Cite, don't over-claim.** Two claims
> were *killed* in verification ? recorded in §5.5 so we don't repeat them. **License caveat:** the
> code repos below are research releases; **independently confirm each repo's actual license is
> MIT/Apache-2.0/BSD before vendoring any of it** (this survey did not verify every license file).
> **Governance:** none of these techniques require training on paid-LLM outputs ? all are compatible
> with our no-Gemini-training-data guardrail (§8).

5.1 Windowing & boundaries (TAL/TAS) ? the explicit-boundary machinery we lack
The recurring lesson: modern action-localization models have an **explicit boundary mechanism**; our heuristic has none (it merges on a velocity+gap rule). The transferable *idea* is "predict a boundary and cut there," but the deep models are built for **hundreds+ of training videos**, so at n=6?16 they are **research bets**, not drop-ins.

- **ASRF ? Action Segment Refinement Framework** (Ishikawa et al., WACV 2021, [arXiv:2007.06866](https://arxiv.org/abs/2007.06866)).
  A shared feature extractor feeds **two branches**: frame-wise classification + a **Boundary Regression Branch**; predicted boundaries refine/split the segmentation (+13.7% edit, +16.1% segmental F1 over MS-TCN-era SOTA on 50Salads/GTEA/Breakfast). **Why it fits:** the boundary head is exactly the missing "cut here" mechanism. **? Two caveats:** (1) ASRF's headline goal is to *reduce* over-segmentation (merge spurious fragments) ? the opposite of our over-*merge*; only the *split-at-a-detected-boundary* sub-mechanism transfers. (2) The claim that ASRF's

boundary branch is a **model-independent post-hoc splitter** you can bolt onto any dense signal was **REFUTED** (§5.5) ? you must train the two-branch model, not graft the branch. **Effort/payoff:** research bet.

- **TriDet** (Shi et al., CVPR 2023, [arXiv:2303.07347](https://arxiv.org/abs/2303.07347)). A **Trident-head** models each boundary as a *relative probability distribution over local bins* (boundary = expected value over neighboring instants) ? built for imprecise/ambiguous boundaries. 69.3% avg mAP on THUMOS14 (> ActionFormer 66.8% at ~75% latency), **convolution-only (SGP), no self-attention** (attractive on a single local GPU). Ablation shows the Trident-head's gain concentrates at strict tIoU=0.7 ? it buys *boundary precision*. **Effort/payoff:** research bet (trained on 100s of videos); the portable piece is the relative-boundary idea.

- **ActionFormer** (ECCV 2022, [arXiv:2202.07925](https://arxiv.org/abs/2202.07925)). The reference single-stage anchor-free template: multiscale pyramid + local self-attention; classify every moment + regress its boundaries (71.0% mAP@0.5 THUMOS14). This is the family TriDet/TemporalMaxer/ CausalTAD extend, and the natural target if we ever graduate from heuristic+LogReg. **Research bet.**

- **TemporalMaxer** (CVPRW 2023, [arXiv:2303.09055](https://arxiv.org/abs/2303.09055)) ? **the most important small-data signal in this dimension.** It replaces ActionFormer's self-attention block with a **parameter-free max-pool** and *beats* it (67.7 vs 66.8 avg mAP) at **~4× fewer params**. The ablation is the takeaway: a **learned 1-D conv block is the WORST** (59.4, overfits redundant features); parameter-free pooling wins. **Why it fits:** direct evidence to **keep our scorer parameter-light** (the numpy LogisticRegression over a handful of features is the *right* choice at our scale, not a placeholder to rush past). **Effort/payoff:** a discipline (free) ? adopt the principle, not the model. *(Caveat: the overfitting evidence is about feature redundancy on a mid-size set, a directional inference to our small-n setting.)*

- **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ? *steering-away note.* Causal-attention masking (attend only to past/present) won EPIC-Kitchens/Ego4D 2024 tracks, but it adds **no** boundary-split module. Useful later for *online/streaming* rally detection; **NOT** the over-merge fix. Listed so we don't mistake causal attention for the remedy.

- **Metrics (the actionable cross-cut):** the field leads with **segmental Edit (Levenshtein) score and F1@k**, not frame accuracy (MoF) or plain tIoU ? see §5.4.

5.2 Shuttle-trajectory ? rally signal processing ? published badminton systems already do what we lack
This is the highest-confidence, lowest-effort dimension: in-domain papers operationalize the exact boundary heuristics we're missing, in spirit license-compatible (re-confirm before vendoring code).

- **See et al. ? Service & End-of-Rally Detection in Badminton** (2024 Springer chapter [10.1007/978-3-031-69073-0_15](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15); 2025 SN Computer Science follow-on [10.1007/s42979-025-03935-0](https://link.springer.com/article/10.1007/s42979-025-03935-0)). **(a) Rally END = a sustained-inactivity rule:** the rally ends when **?80% of consecutive frames show no shuttle motion** (~**95% accuracy** over 187 rallies). **(b) Rally START (service) = a learned classifier** over court geometry + player poses (~26 features/frame × 12 frames ? CNN/BiLSTM, **88.6%** accuracy). **Why it fits:** the inactivity rule is *precisely* the end-of-rally cut our windowing lacks ? a few lines, directly breaks 65 s mega-windows. **? Caveats:** the rule is a *termination*, not a *within-rally split*; results rest on paywalled abstracts + small test sets (35?187 rallies); the serve classifier needs the (default-OFF) person cue + court polygons. **Effort/payoff:** the inactivity rule = **low-effort high-confidence win**; serve classifier = medium.

- **Hsu, Yu & Cheng ? trajectory + swing fusion** (Sensors 2024, 24(13):4372, [10.3390/s24134372](https://www.mdpi.com/1424-8220/24/13/4372)). Fusing TrackNet trajectory with a YOLOv7 swing detector via a Shot Refinement Algorithm raises hit detection **F1 72.3% ? 90.5%** (over 69 BWF matches / 1582 rallies). Hits are found by **trajectory direction-reversal / peak detection** (convert (x,y)?(frame,y) wave peaks) then reconciled with the swing cue. **Why it fits:** validates our **late/score-level fusion** approach in-domain, and direction-reversal/peak is a cheap **per-window feature** marking contact events (a "rally has hit cadence" signal ? the discriminator a "shuttle-moving" heuristic lacks). **? Important caveat (the one 2-1 vote):** this paper does *within-rally stroke/hit* segmentation, **not** rally start/end ? "it splits merged rallies" is *our extrapolation*; frame it as "hit cadence plausibly helps locate rally boundaries," not a proven rally-splitter. **Effort/payoff:** the peak/direction-reversal feature = low-effort win (feed the scorer); full swing fusion = medium.

- **What best separates "rally" from "shuttle merely moving"** (synthesis): sustained-inactivity termination (See), hit cadence / direction-reversals (Hsu), net-crossings ?2 (our Gate-A, [`rally_gate.py`](../backend/pipeline/detectors/rally_gate.py)), two-sided court occupancy (our person features). None alone is sufficient; the spine (§4) says **feed them all to the learned scorer**.

5.3 Small-data strategies ? most quality per label
- **Timestamp / sparse supervision** (Li et al., CVPR 2021, [arXiv:2103.06669](https://arxiv.org/abs/2103.06669); TAS survey Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352)). Label **one frame per action** ? iteratively refine pseudo frame-labels ? *comparable* to full supervision (50Salads: timestamp 75.6% acc / 60.1 F1@50 vs full 83.7% / 70.1). **Why it fits:** sparse per-

rally timestamps cost far less than dense boundary labels ? relevant if we ever train a frame-wise model. **? Caveats:** "comparable" hides a real **~8?10 pt residual gap**; annotation savings are modest (~35%, annotators still watch the whole clip); benchmarks are cooking/egocentric ? transfer to 1-D shuttle trajectories is **unproven**. **Effort/payoff:** research bet to pilot.
- **Parameter-light scorer** ? see TemporalMaxer (§5.1): keep the numpy LogisticRegression; resist a learned temporal net until the corpus is much larger. **Low-effort win (a discipline).**
- **? OPEN (the research could NOT verify these ? honest gap):** *active learning* (which video/segment to label next via uncertainty/diversity/core-set) and the *teacher-student / distillation governance boundary* for using Gemini as a weak/pseudo-labeler returned **no surviving verified claim**. Our pragmatic, guardrail-safe stance (not from the literature ? from our own design): pick the next video to label by **maximal local-vs-Gemini disagreement** (the §4.4 shadow signal) + **scorer uncertainty** + **venue/fps diversity**; and treat **Gemini strictly as an inference-time weak-reference / eval oracle ? never a training-data source for owned weights** (§8). These two topics deserve their own grounded research pass before we lean on them.

5.4 Evaluation rigor at small n ? lead with segmental metrics
- **Frame accuracy (MoF) and plain tIoU are BLIND to over-segmentation** (TAS survey, Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352); segmental-F1 origin Lea et al., CVPR 2017). "The score can be high even when segments are fragmented." The over-merge that bit us is exactly what tIoU-F1 hides (a giant window still *overlaps* real rallies). **Action:** lead the leaderboard with the **segmental Edit (Levenshtein) score** (penalizes spurious ordering/fragmentation) **and F1@{0.1,0.25,0.5}**, paired with our existing `segment_count_ratio` + the merge/split breakdown. **We already have** F1@k, mAP@tIoU, segment-count-ratio, bootstrap CIs, paired sign-test, Platt/ isotonic + ECE/Brier ? the only *missing* primary metric is the **Edit score** (low-effort add).
- **? OPEN (not covered by a verified claim):** IAA / label-noise level in the golden set, and which over-segmentation-robust loss (ASRF boundary/smoothing, MS-TCN truncated-MSE) best tolerates it at small n. Measure IAA as the corpus grows (Roora has an A/A overlap plan); revisit robust losses only when a frame-wise model is on the table.

5.5 Refuted claims (recorded so we do NOT repeat them)
1. **"ASRF's boundary branch is model-independent ? a post-hoc splitter applicable to any dense segmentation output."** **REFUTED (0-3).** Implication: we cannot bolt ASRF's boundary head onto our trajectory signal as a drop-in; the two-branch model must be trained together. ? port the *concept*, don't expect a plug-in.
2. **"A GFL-inspired Boundary Distribution Regression head is the mechanism most relevant to splitting merged windows."** **REFUTED (0-3)** (and the cited source had a suspect identifier). Do not adopt

this framing as established.

5.6 External due-diligence review (2026-06-17) ? owner-commissioned; verify before submission

An owner-commissioned external review (31 cited sources) of the [`RALLY_DETECTION_QUALITY_REPORT.md`](RALLY_DETECTION_QUALITY_REPORT.md) checkpoint **validated the methodology** (the decoupled 2-layer design, the R2 metric, LOVO + paired-sign-test discipline, the consented-data + over-seg-aware-harness moat) and surfaced these **borrowable inputs**. These are *research inputs for later pickup* ? they add **no measured numbers**, and the citations are the review's own and **must be re-verified before any submission**.

- **Positioning ? the "first-mile" gap (highest-value).** Badminton stroke-classification SOTA (**BST** ? Badminton Stroke-type Transformer, CVPR-W 2026; **DSTA** ? Decoupling Spatio-Temporal Adapter) and the big datasets (**ShuttleSet** ~36,492 strokes; **Fine-Badminton** ~27,597 actions) all **silently assume a pre-trimmed rally clip already exists** ? they depend on a human having segmented the untrimmed match. Our system solves that ignored *prerequisite*. Reframe the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage"** ? a real, unoccupied gap and a far stronger narrative than "applied/systems."
- **R2 ? Action-Spotting / Precise-Event-Spotting (AS/PES).** Our asymmetric tolerance-match metric sits at the **TAL × AS intersection** (interval task, tolerance-based match). Grounding it in the AS/PES literature (SoccerNet, TenniSet, T-DEED, the PES survey [arXiv:2505.03991]) frames R2 as a deliberate methodological contribution, not an ad-hoc metric ? strengthens §5.4 and the report's §9.
- **Comparative baselines reviewers will require.** Benchmark against **MonoTrack** and **BST** on **ShuttleSet / Fine-Badminton** ? we currently have only a generic heuristic + the Gemini wrapper; domain-specific baselines are a hard publication requirement.
- **Scorer upgrade ? prefer TemporalMaxer.** For the post-logreg scorer (corpus-gated; NEXT_STEPS M0.4), **TemporalMaxer** (parameter-free temporal max-pool; ~2.8× fewer GMACs; robust at small-N, [arXiv:2303.09055]) fits the cheap/local/fast mandate better than attention-heavy ActionFormer.
- **Court polygons ? pose ? the rally-END lever.** Court masking is not only the person-cue spectator fix (§4 / `MULTI_SIGNAL_FUSION_PLAN.md`): once pose is isolated to on-court players it can resolve **ambiguous, occluded end-of-rally states** ? feeding directly into the **#1 remaining gap (the rally-END boundary)**, not just precision. Connect court-polygons to the R4/END-trim work.
- **Name the eval?serve lesson "pipeline distribution skew."** Downstream-cue value depends on the upstream proposal distribution (the Gate-A episode) ? a citable framing and candidate paper "lesson."
- **Citable Layer-1 anchor (public benchmark, NOT our footage).** WASB SBDT F1 95.6 (step=1) / 94.0 (step=3) vs MonoTrack 92.1, TrackNetV2 89.4, BallSeg 71.7 (WASB, BMVC 2023). Justifies the frozen detector choice; **keep distinct from our own, still-unmeasured footage domain-gap** (§2.1, a real

open limitation the review also flags).

Sequencing: these are *publication-prep + post-current-idea* levers. The prerequisites remain corpus growth to n?16 and **multi-court isolated as its own reported slice** (§6 Tier-3 / §7) ? do those first; the items above land once current windowing/fusion ideas are exhausted.

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6. Prioritized roadmap (cheap ? durable)

Every item is **its own measured PR**: default-OFF, scored on the §4 metric set over the golden set with honest stats (CI + sign-test), **re-confirmed on the served windowing** (the eval?serve gate), logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md). Ordered by the research's effort/confidence tiering. Each carries a **falsifiable success criterion** ? if it isn't met, the item does not promote (and that null is itself recorded).

Definition of "bridging the gap"
Local rally segmentation, measured **honestly LOVO-by-match on the golden set**, approaches the Gemini reference (today **0.93 single-court / 0.66 multi-court** tIoU-F1) ? but tracked primarily through the **over-segmentation-sensitive metrics** (segmental Edit + F1@k + low merge/split rate), since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is rising **agreement-with-Gemini** on the unlabeled pool (§4.4, shadow only).

Tier 0 ? Make the failure measurable FIRST (do before any fix)
- **E1 · Over-segmentation-aware leaderboard.** Add the **segmental Edit (Levenshtein) score** to `segmentation_metrics.py`; surface Edit + F1@{0.1,0.25,0.5} + `segment_count_ratio` + a **merge-rate / split-rate** breakdown (promote the `scratch/compare_local_vs_gemini.py` matching into a reusable `backend/eval/` helper) in the default `experiment.compare()` report; add the one-command rally-segmentation eval entrypoint (§4.5). **Success:** the current heuristic's over-merge shows up as a *number* (low Edit, high merge-rate) on the 6 golden videos, and a baseline row lands in the leaderboard. *(Low effort, high confidence ? research-backed §5.4.)*

Tier 1 ? Low-effort, high-confidence wins
- **W1 · Bounded windowing (the direct over-merge fix).** Add a `max_window_duration` cap and a **sustained-inactivity split** to `trajectory_to_action_windows` ? port See et al.'s rule (split / end a window when ?~80% of consecutive frames over a trailing window show no shuttle motion). Config-gated, default preserves current behavior. **Success:** segmental Edit + F1@k improve vs `CONTROL` on the golden set (?F1 CI excludes 0 **and** sign-test agrees) with **no served-windowing regression**; merge-rate drops materially. *(Win ? §5.2 See et al., ~95% end-of-rally acc.)*
- **W2 · Rally-state features into the learned scorer (no special-casing).** Wire the already-built, default-OFF cues as scorer feature columns: **net-crossings** (Gate-A), **two-sided motion**, and a

new cheap trajectory feature ? direction-reversal / peak count (Hsu) as a hit-cadence proxy; audio hit-rate when the hit detector lands. **Re-measure with W1 in place and on the served windowing** ? recall person was ?0.021 and audio +0.079 but *both CIs spanned 0* at n=6 on the *pre-fix over-merged* candidates; the picture may change once windows are bounded. **Success:** the ablation (`ablation.py`) shows a cue with ?F1 CI clearing 0 + sign-test, surviving served re-check ? promote that cue; others stay default-OFF + retained (no deletion on a small-n null). *(Win/medium.)*
- **W3 · Keep the scorer parameter-light (a discipline, documented not coded).** Per TemporalMaxer's ablation (learned temporal conv overfits; parameter-free wins), resist replacing the numpy LogisticRegression with a learned temporal net until the corpus is much larger. Record this as a standing decision. *(Win ? §5.1 TemporalMaxer.)*

Tier 2 ? Medium effort
- **M1 · Serve-start classifier ? rally-start boundary** (See et al. pose/court, 88.6%). Needs the person cue + a **court-mask source** (manual polygon vs auto-homography ? *owner-gated* per [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §9.1). **Success:** start-boundary timing error drops on golden; promotes only if it lifts segmental F1@k without served regression.
- **M2 · Per-venue threshold calibration** against golden labels (the heuristic's footage-fragility ? it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local segment counts track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0 head's 0.561 without harming clean-court F1. *(Velocity is already fps-normalized ? calibrate per-venue, not per-fps.)*

Tier 3 ? Research bets (durable; corpus-gated)
- **R1 · Owned segmenter with an explicit boundary branch** (the ASRF/TriDet *idea* ? a boundary head that predicts cut points ? ported into our owned TAL head, NOT the full deep model, and **trained**, not grafted (§5.5 refutation)). Per [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md): **TAL head first, VLM later.** **Gate:** the central unverified assumption is whether boundary-head gains hold at our scale on trajectory features ? so **pilot only when the corpus is materially larger (~?16?30 matches)**, and only after W1/W2 establish the heuristic ceiling. **Success:** beats the tuned heuristic + LogReg on held-out matches (segmental Edit + F1@k), honestly LOVO.
- **R2 · Timestamp/sparse supervision** to get more labeled rallies per annotator-hour (§5.3) ? pilot if/when we train a frame-wise model; research bet (residual gap, unproven on trajectories).

Throughout ? corpus growth + active learning + label quality
- **Grow the golden set 6 ? ~16** via Roora ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)) ? the single highest-leverage lever ([`NEXT_STEPS.md`](NEXT_STEPS.md)).
- **Active learning:** select the next videos to label by max **local-vs-Gemini disagreement** (§4.4)

  + scorer uncertainty + venue/fps diversity (our pragmatic stance; §5.3 flags the formal methods as
  an open research pass).
- **Measure IAA / label noise** as the corpus grows (Roora A/A overlap); revisit over-segmentation-
  robust losses only when a frame-wise model is on the table (§5.4 open question).

Dead-ends / do-NOT (so we don't relitigate)
- Don't treat **causal attention** (CausalTAD) as the over-merge fix ? it has no boundary-split (§5.1).
- Don't expect **ASRF's boundary branch as a post-hoc drop-in** on our dense signal ? refuted (§5.5).
- Don't adopt a **learned temporal net** at n=6?16 ? TemporalMaxer evidence says it overfits (§5.1).
- Don't chase **bare tIoU-F1 / frame accuracy** ? blind to the over-merge (§5.4).
- Don't let **Gemini outputs become training data** for owned weights ? inference/shadow only (§8).

---

7. Project: plan, tracking & definition of done

This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection gap."** It
completes when all committed tiers land AND a single quality number is attributed to the effort,
measured honestly and logged. This §7 is the living burn-down (update the status column as PRs merge;
log each result in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), numbers in
`eval_baselines/experiment_leaderboard.json`).

7.1 The number (what we attribute to this effort)
Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ?~16):
- **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus **gap-closure %**
  = `(final ? baseline) / (Gemini ? baseline)`. Today: heuristic **0.157** ? Gemini **0.66**
  (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ? 50% of the
  heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally beating Gen-0's 0.561.
- **Primary lens (over-segmentation-sensitive, the actual failure):** segmental **F1@0.25** +
  Edit-score + **merge-rate** ? these must improve even if tIoU-F1 moves little.
- **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only if it also
  holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1** before any fix.
- **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit, not a single
  video) via `python -m backend.eval.rally_seg_eval`. The 0.157/0.66 figures above are per-venue
  references; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the *same* golden
  set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).

Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):

after	F1	merge_rate	? vs baseline
E1 baseline (current heuristic)	0.300	0.523	?
+ W1 bounded windowing (cap=12, **default-OFF**)	**0.439**	**0.038**	**+0.139** (6/0 sign-test)
+ W2 net-crossings gate (mc=1, **default-OFF**)	**0.457**	0.038	**+0.157 cumulative** (W2 step +0.019, 6/0)

+ W1.5 hard-cap fallback (**default-OFF**)	0.444	0.000	**+0.006 (n.s., CI spans 0)** ? neutral on golden; real-footage fix (mahadevpura-1 174s blob ? 24×?9s windows)
+ W2 multi-feature scorer (person/audio)	_next (GPU)_		

7.2 Work items (status checklist)

Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured PR.)

ID	Deliverable	Tier	Depends on	Acceptance / promote criterion	Status
E1	merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185)	0	?	over-merge shows up as a number on the 6 golden videos	? baseline F1 0.300, merge_rate 0.523
W1	Bounded windowing ? `max_window_duration` cap + inactivity split (See et al.) (#187)	1	E1	seg F1@k ? vs baseline (CI?0 **and** sign-test), no served regression; merge-rate ?	? **?F1 +0.139 (6/0), merge_rate 0.523?0.038** (default-OFF)
W1.5	Hard-cap fallback (`window_hard_cap`) ? chop continuously-active windows at the cap when no inactivity gap exists	1	W1	golden ?F1 ?0 (or neutral) **+** fixes a real-footage over-merge it can't otherwise cap	? **golden ?F1 +0.006 (n.s.)**; real `mahadevpura-1` 174s blob ? 24×?9s; retained **default-OFF** (real-footage + true-cap rationale, not a golden lift)
W2	Rally-state features (net-crossings ? #188 + **never-empty safeguard** ?; two-sided motion, **direction-reversal/peak**, audio = follow-on)	1	E1, W1	?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF	? **net-crossings ?F1 +0.019 (6/0)**, gate now **do-no-harm** (never zeroes a video); multi-feature scorer next (GPU)
W3	Keep scorer parameter-light (standing decision; TemporalMaxer evidence)	1	?	documented; no learned temporal net until corpus ? now	?
M1	Serve-start classifier ? rally-start boundary (pose/court)	2	W1, **court-mask decision (owner-gated)**	start-boundary timing error ?; promote on seg-F1 lift, no served regression	?
M2	Per-venue threshold calibration vs golden	2	E1	multi-court F1 ? toward Gen-0 0.561 without harming clean-court	?
R1	Owned segmenter with an explicit **boundary head** (port ASRF/TriDet *idea*, trained ? not grafted)	3 (stretch)	corpus ?~16?30, W1/W2	beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO	?
R2	Timestamp/sparse-supervision pilot (label-efficiency)	3 (stretch)	R1 path	measured label-efficiency win	?
C	Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA	ongoing	?	corpus ?16; IAA recorded	? (~6 now; ~10 by the weekend)

7.3 Sequencing

`E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2` (corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage input and gates the Tier-3 bets.

7.4 Definition of done

The project is **complete** when:

1. **Tier 0 + Tier 1 + Tier 2** work items are merged ? each a measured PR (default-OFF, promoted only on measured lift, re-confirmed on the served windowing).
2. **The headline number (§7.1) is recorded** in `QUALITY_ITERATIONS.md` + the leaderboard, with honest stats, on the grown golden set ? **and the gap-closure target is met, or the honest null is recorded with a reasoned go/no-go on Tier 3.**
3. Each landed item left the harness greener (no regression) and `QUALITY_ITERATIONS.md` tells the story.

> **Tier-3 scope decision (owner-adjustable):** R1/R2 (the owned boundary-head model + sparse
> supervision) are **corpus-gated stretch goals** ? the verified research says deep TAL detectors risk

```
> overfitting at n=6?16. **Default:** the project *completes* at Tier 0?2 + an attributed number, and
> Tier 3 spins out into the owned-model milestone ([`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).
> Fold Tier 3 into the DoD instead if you want the project to span the owned model.

---

## 8. Constraints & guardrails (non-negotiable) + cross-links

- **Licensing:** every dependency ? code, data, AND weights ? must be **MIT / Apache-2.0 / BSD**.
  Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).
- **Never train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (terms/copyright);
  paid LLMs are inference/serving/shadow-reference only. Also forbidden: public datasets known to
  be LLM-generated. **Exclude minors** from any training pool; per-user data needs explicit opt-in.
- **Owned model:** TAL head FIRST (de-risked, trains in minutes, doubles as a pseudo-label teacher +
  honest baseline the VLM must beat), then Qwen3-VL (Apache-2.0) / InternVL3 (MIT) ? see
  [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
- **Cross-links:** [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) ·
  [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) · [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) ·
  [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) · [`ABLATION_LAYER.md`](ABLATION_LAYER.md) ·
  [`EVALUATION.md`](EVALUATION.md) · [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) ·
  [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) · [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
```

6. user (2026-06-20T08:00:36.515Z)

```
# Request-Access form ? segmented lead capture + setup discovery

> **Status:** `IN PROGRESS` · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
> **Relation:** first tracked work item in the `khelsutra` repo; mirrors the engine repo's
> `PROJECT_DOC_TEMPLATE` discipline (status header · decisions-locked · §6 tracker · DoD).
> Feeds the platform direction in the engine repo's `PLATFORM_ARCHITECTURE.md §3` (StorageBackend /
> ComputeBackend) and `VIDEO_LOCALITY_MODEL.md` ? see §7.

---

## 0. TL;DR
Replace the landing page's `mailto:` "Request access" CTA with a real form that (a) segments
**academies vs individual enthusiasts**, (b) asks the two questions that actually decide how we'd
serve a user ? **"do you have a GPU machine?"** (compute placement) and **"where do your videos
live?"** (storage backend) ? and (c) keeps a free-text message. Submissions are stored in our own
**Cloudflare D1** via a small **Worker** (`POST /api/request-access`). The form doubles as PMF +
architecture research.

## 1. Problem & goal
Today the CTA is `mailto:hello@khelsutra.guru` ? zero structure, nothing collected, no
```

segmentation.
We want **structured, queryable** interest signals from two distinct audiences, captured the moment someone raises a hand, **without shipping lead data to a third party** (privacy-first ethos).
Goal: a clean on-brand form whose responses land in our own Cloudflare account, queryable later.

2. Decisions locked
- **Backend = Cloudflare Worker + D1** (owner decision, 2026-06-20). Data stays in our Cloudflare account; queryable; no third party. (Alternatives considered: hosted form (Tally/Formspree ? third-party data), Worker?email-only (no DB), structured `mailto` (no collection). Rejected for the reasons in parens.)
- **Two audience segments:** `academy` and `individual` ? the form lightly adapts (academy also asks club name + #courts).
- **The two discovery questions are first-class** (not optional polish): GPU/compute + storage ? they map 1:1 onto the BYO-compute / BYO-storage architecture forks (§7).
- **Free-text message retained** ("Anything else?").
- **Privacy:** store only what's submitted + a coarse country (from `request.cf.country`); **no raw IP**; consent checkbox for "email me about the alpha"; this data is **operations/contact only**, never a training source (same guardrail as the engine repo).
- **Deploy model changes:** the site moves from dashboard direct-upload ? **Wrangler** (Worker with static assets + D1 binding). Documented in §5; owner runs the `wrangler` deploy (our account/auth).

3. Form spec
Field	Type	Notes
Segment	radio	`academy` | `individual` (required)
Name	text	required
Email	email	required (so we can reply)
City	text	optional
Club / academy name	text	shown for `academy`
Courts recorded	select	`1` | `2-4` | `5+` (academy; optional)
GPU machine for local processing?	radio	`yes` | `unsure` | `no` | `prefer-cloud`
Where do your recordings live / how would you share them?	checkboxes (multi)	`gdrive` | `onedrive` | `dropbox` | `bucket` (S3/GCS) | `local` (SD/drive) | `unsure`
Message	textarea	free text ("Anything else?")
Consent	checkbox	"Okay to email me about the KhelSutra alpha"
`hp`	hidden honeypot	anti-spam; if filled ? silently drop

4. Design / architecture
- **Frontend** (`site/index.html`): the header + hero "Request access" buttons scroll to the `#join` form; `#join` becomes the form. On submit, JS `POST`s JSON to `/api/request-access`. **Progressive enhancement:** if the endpoint isn't live yet (pre-Worker), it falls back to a prefilled `mailto:hello@` so the form is useful from the first re-upload.
- **Worker** (`/api/request-access`): validates (email present, honeypot empty), inserts into D1, returns `200 {ok:true}`; serves the static assets for all other routes.
- **D1 schema** (`access_requests`): `id, created_at, segment, name, email, city, org_name, courts, has_gpu, storage (csv), message, consent, country, user_agent`.
- **Anti-spam:** honeypot first; add **Cloudflare Turnstile** in P2 if spam appears.

- **Notify (P2):** optional email ping to `hello@` per submission (MailChannels / Cloudflare Email).

5. Deploy (owner-run; our Cloudflare account)
```
wrangler d1 create khelsutra-access            # once; copy the database_id into wrangler.toml
wrangler d1 execute khelsutra-access --file=site/schema.sql   # create the table
wrangler deploy                                # ships the Worker + static assets + D1 binding
```
> Until the Worker ships, the form works via the `mailto` fallback (re-upload `site/` as today).
> Git?Cloudflare auto-deploy still intentionally not connected.

6. Deliverables & progress tracker ? source of truth
Legend: ? Todo · ? In progress · ? Done. One PR per row off `origin/master`.

ID	Deliverable	Gated?	Status	PR
P0	This design doc + `NEXT_STEPS.md`	No	?	#2
P1	Form UI (segments + GPU + storage + free text + consent) w/ `mailto` fallback	No	?	#2
P2	Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)**	owner deploy	?	this PR (built+tested; owner runs `wrangler d1 create` + `deploy`)
P3	Anti-spam (Turnstile) + per-submission email notify	No	?	?
P4	Tiny admin/export view (auth'd) to read responses	owner	?	?

7. Why this matters beyond lead capture
The two questions are **the architecture forks** from the engine repo's BYO analysis:
- **GPU?** ? compute placement: desktop/local (their GPU) vs hosted vs zero-infra-BYO
  (`PLATFORM_ARCHITECTURE.md §3.2`, `DESKTOP_APP_PLAN.md` P0 spike).
- **Storage?** ? which `StorageBackend` + whether autosync is worth building
  (`PLATFORM_ARCHITECTURE.md §3.1`, the cloud-storage/autosync thread).
So early responses directly inform what to build next ? not just who to email.

8. Open questions
1. Turnstile now or wait for spam? (Default: honeypot now, Turnstile if needed.)
2. Email-notify per submission, or just query D1 weekly? (Default: notify in P2.)
3. Do we localize the form (Kannada/Hindi) for academies now or at beta? (Default: English now.)

9. Definition of done
- [ ] Form replaces the `mailto` CTA, segments academy/individual, asks GPU + storage + free text.
- [ ] Submissions persist to Cloudflare D1 via the Worker; honeypot drops bots.
- [ ] Privacy posture honored (no raw IP; consent; contact-only).
- [ ] Deployed live on khelsutra.guru; a test submission appears in D1.
- [ ] Doc tracker + DoD updated; archived when P1?P3 land.

7. user (2026-06-20T08:00:36.552Z)

Portability ? Cloudflare now, always-on VM if needed

> **Status:** design + proven (2026-06-20). **Decision:** keep the site/form host-portable so
> moving off Cloudflare to an owner-run always-on VM ("if Cloudflare acts out") is an **adapter
> swap, not a rewrite.**

The shape that makes it portable
- **Logic is runtime-agnostic.** `src/index.js` uses Web-standard `Request`/`Response` and pure
  helpers (`validate`, `toRow`, `isBot`, `handleRequest`) ? no host-specific calls.
- **All DB coupling is in adapters** (`src/store.js`): `d1Store` (Cloudflare D1) and

```
`sqliteStore`
  (node:sqlite / better-sqlite3). `handleRequest(request, store)` takes the store by injection.
- **One schema, one SQL.** `schema.sql` and the `INSERT` are plain SQLite, so D1 and a VM's SQLite
  file are byte-for-byte compatible ? zero dialect change. (Postgres later = minor placeholder tweaks.)

## Two hosts, same code
| | Cloudflare (now) | Always-on VM (fallback) |
|---|---|---|
| Entry | `src/index.js` (Worker `fetch`) | `server.js` (`node:http`) |
| Static | `env.ASSETS` | `server.js` serves `public/` |
| Storage | `d1Store(env.DB)` ? D1 | `sqliteStore(db)` ? `node:sqlite` file |
| Country | `cf-ipcountry` header | optional (GeoIP / null) |
| Deploy | git ? Workers Builds | `node server.js` under systemd |
| TLS/DNS | Cloudflare | keep Cloudflare in front, **or** Caddy/Let's Encrypt |

**Enforced, not asserted.** `node --test src/` runs BOTH paths and CI (`.github/workflows/test.yml`,
Node 24) runs them on every PR, so the repo stays compatible with both hosts:
- `src/index.test.js` ? the Cloudflare Worker + D1 logic.
- `src/server.test.js` ? **boots the real `server.js`** (node:http + node:sqlite) and round-trips it
  over HTTP into SQLite (the same proof we run on the droplet, automated).

For a live check against a real VM: `SSH_TARGET=root@host scripts/droplet-smoke.sh` (scp + boot +
round-trip + cleanup). Verified live on the owner's droplet (Ubuntu 24.04, isolated `/srv/khelsutra-site`).

## VM deploy runbook ("if Cloudflare acts out")
Node 22+ only ? **no `npm install`** (node:sqlite is built in).
```
```
# on the VM:
git clone ? && cd khelsutra/site        # or scp the site/ folder
PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
# put it under systemd (auto-restart) + a reverse proxy (Caddy/nginx) for TLS,
# then point DNS at the VM. ~1 hour; logic + schema unchanged.
```
```

## Honest notes
- "Cloudflare acts out" usually means **Workers/D1 pricing or limits, not DNS** ? so the cheapest
  move is to keep Cloudflare DNS/CDN in front and shift only the **compute** to the VM (point a
  route/origin at the box). Full exit (own DNS + Caddy TLS) is also supported.
- You can run **both** (Cloudflare primary + VM warm standby) since the schema is identical; a
  periodic `access_requests` export keeps them in sync.
- `server.js` is intentionally dependency-free; if you later want Postgres or a framework (Hono runs
  on Workers *and* Node), it slots in behind the same `store` + handler.
```

8. assistant (2026-06-20T08:00:39.647Z)

I have everything I need from the canonical template and the three real-world example docs. Let me compile the exact schema.

9. user (2026-06-20T08:01:27.691Z)

Structured output provided successfully

10. assistant (2026-06-20T08:01:32.191Z)

I've extracted the exact tracked-design-doc structure and returned it via StructuredOutput. The schema is grounded in the canonical template (`docs/PROJECT_DOC_TEMPLATE.md`) and verified against three real in-use examples (`RALLY_QUALITY_RESEARCH.md`, `PROMINENT_COURT_DETECTION.md`, and the khelsutra `REQUEST_ACCESS_FORM.md` / `PORTABILITY.md`), with path:line citations for every field.